Towards Reliable Multi-KB ToDs: Improving Task-Oriented Dialogue Systems in Multiple Knowledge Bases Through Multi-Agent Framework

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Abstract

End-to-end task-oriented dialogue systems in Multi-Knowledge Bases predominantly rely on resource-intensive annotated datasets, which are difficult to obtain in real-world scenarios. Large language models' application in this field can effectively alleviate the problem of data scarcity. However, directly applying LLMs to these tasks frequently results in hallucinations and inconsistencies due to complex multi-turn constraints. To address these limitations, we propose MAMBA, a Multi-Agent System for multi-KB ToD. This framework eliminates the need for labeled training data by decomposing the dialogue process into three collaborative agents: a ***Sketch Planning Agent*** for sketch response generation, a ***Knowledge Retrieval Agent*** for precise knowledge retrieval, and a ***Final Response Agent*** for final response synthesis. Furthermore, we introduce an iterative reflection mechanism that evaluates and refines intermediate outputs to ensure factual groundedness. Extensive experiments on the CrossWOZ and RiSAWOZ benchmarks demonstrate that our training-free framework significantly outperforms the state-of-the-art supervised baseline, KoK-HAN. We hope that our research offers a valuable resource for developing training-free and reliable dialogue systems.

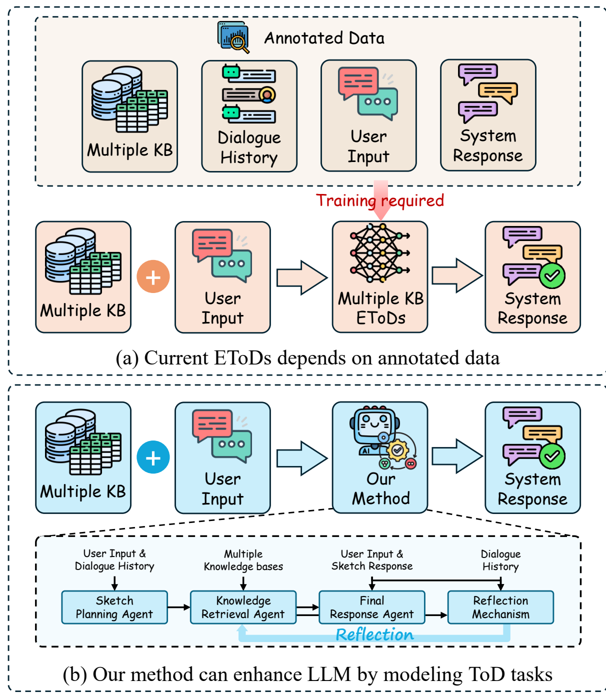


Figure 1: End-to-end task-oriented dialogue system requires annotated data for training (a). Our method improves the performance of LLMs by effectively modeling the ToD tasks, without relying on training data (b).

1 Introduction

Task-oriented dialogue systems (ToDs) [Young *et al.*, 2013] serve as critical interfaces for human-computer interaction, aiming to assist users in completing specific goals, ranging from technical support to complex itinerary planning. Unlike open-domain conversational systems, these systems are required not only to understand user intent precisely but also to ground their responses in factual information. This challenge is especially evident in multiple knowledge bases (multi-KB) scenarios [Qin *et al.*, 2023a], where data heterogeneity and complexity increase significantly.

With the advancement of deep learning, researchers have primarily utilized end-to-end task-oriented dialogue systems (EToDs) to address these tasks [Zhang *et al.*, 2020; Balaraman *et al.*, 2021; Qin *et al.*, 2023b]. For instance, Zeng *et al.* [2022] propose HM2Seq, which enhances retrieval accuracy through hierarchical memory and reasoning. Wan *et al.* [2023] introduce a multi-grained knowledge retriever to decouple retrieval from generation, benefiting performance in large-scale knowledge bases. Qin *et al.* [2023a] present KoK-HAN, a multi-connection network specifically modeled for multi-KB scenarios. More recently, applying Large Language Models (LLMs) to ToDs has become a key research direction [Qin *et al.*, 2023b]. Dong *et al.* [2025] propose a framework enhancing LLM performance through adaptive exploratory retrieval, while Rim *et al.* [2025] introduce CT-Fusion for evaluating production systems.

Despite these advancements, significant limitations persist in ToDs for multi-KB: **Dependency on Annotated Data**. Current end-to-end task-oriented dialogue systems depend heavily on fully annotated datasets, which is illustrated in Figure 1 (a). This indicates that the current method relies on a substantial amount of data to achieve satisfactory performance, which can be challenging to obtain in real-world scenarios where data is limited. Large language models (LLMs) have been widely proven to have excellent task generalization capabilities [Qin *et al.*, 2024; Naveed *et al.*, 2025; Wang *et al.*, 2025b; Chen *et al.*, 2025], and applying LLMs to data-scarce scenarios in ToD is a potential solution. However, direct application of LLMs often suffers from hallucination and inconsistency when navigating complex, multi-turn ToD tasks [Cui *et al.*, 2025], which makes it difficult for LLMs to achieve sufficient performance.

To bridge these gaps, we propose MAMBA, a **M**ulti-**A**gent system for **M**ultiple knowledge **BA**ses task-oriented dialogue, designed to model multi-KB scenarios without reliance on labeled training data, which is shown in Figure 1 (b). We decompose the multi-KB ToD task into three collaborative roles: the ***Sketch Planning Agent:*** Generate a sketch response that includes placeholders based on the dialogue history and user input, the ***Knowledge Retrieval Agent:*** Search relevant information from multiple knowledge bases according to the sketch, and the ***Final Response Agent:*** Integrate the sketch response with the retrieval information and output the final response. Furthermore, inspired by self-refinement techniques [Madaan *et al.*, 2023], we introduce a reflection mechanism that iteratively evaluates and optimizes the output of the previous stage. This mechanism dynamically evaluates and refines intermediate outputs, ensuring that the final response is both logically consistent and factually grounded, thereby maximizing the reasoning potential of LLMs.

Following previous research [Qin *et al.*, 2023a], we conduct experiments on the CrossWOZ and RiSAWOZ in multi-KB settings. The results indicate that our framework achieves significant improvements, outperforming the state-of-the-art baseline model KoK-HAN without the need for any training. We use the scoring scale established by Li *et al.* [2026] and employ current advanced LLMs to evaluate the response outcomes, demonstrating the effectiveness of our framework in terms of dialogue fluency. We hope this work will offer valuable insights for future research on developing robust, knowledge-based dialogue systems using LLMs.

Our contributions are summarized as follows:

- To the best of our knowledge, this work represents the first attempt to explore the performance of large language models on ToDs under multi-KB settings.
- We propose MAMBA, designed to address this task through precise modeling and collaborative approaches. Furthermore, we introduce a reflection mechanism to improve performance.
- The performance on the CrossWOZ and RiSAWOZ benchmarks further illustrates the effectiveness of our framework. Moreover, it surpasses the carefully designed training framework KoK-HAN without necessitating any additional training.

2 Problem Definition

Unlike single knowledge base ToDs, multi-KB ToDs require the model to reason by integrating information from multiple knowledge bases. The model must first gather the necessary background knowledge relevant to the current user input based on the dialogue history. It then combines information from both knowledge bases and dialogue to derive a final answer and returns the correct response.

Following previous work [Wu *et al.*, 2019; Qin *et al.*, 2020; Qin *et al.*, 2023a], we can formally define multi-KB ToDs as: input multiple knowledge bases $\mathbb{B} = \{\mathcal{B}_1, \mathcal{B}_2, ..., \mathcal{B}_n\}$, dialogue history $\mathbb{H} = \{\mathcal{H}_1, \mathcal{H}_2, ..., \mathcal{H}_m\}$, and user input d_n^i; the model needs to output the response d_n^r based on $\mathbb{B}, \mathbb{H}, d_n^i$. It can be defined as:

$$d_n^r = \text{Model}\left(\mathbb{H}, \mathbb{B}, d_n^i\right), \tag{1}$$

where $\mathcal{H}_m = \left(d_m^i, d_m^r\right)$, d_m^i presents the m-turn user input, d_m^r presents the m-turn system response, and $n = m + 1$. Here, d_n^r contains content retrieved from multiple knowledge bases $\mathbb{B}$. For example, in the response "*I recommend visiting Sight_A*", "*Sight_A*" represents the content extracted from knowledge bases. Finally, this retrieved content will be compared with the slot of ground truth to calculate the F1 score.

3 MAMBA

In this work, we introduce a multi-agent system for multi-KB ToD, which improves the performance of LLMs in the multi-KB ToD task. Firstly, MAMBA comprises three agents: (i) the Sketch Planning Agent, (ii) the Knowledge Retrieval Agent, and (iii) the Final Response Agent, which model the specific steps used in ToDs. Furthermore, MAMBA introduces a reflection mechanism that enhances the performance by evaluating the current results of the ToD modeling. The main framework is illustrated in Figure 2.

3.1 Sketch Planning Agent

The Sketch Planning Agent generates a sketch response d_n^{init} and retrieval request set $\mathbb{S}$ based on the dialogue history $\mathbb{H}$ and user input d_n^i. The sketch response includes special placeholders, $\mathbb{S}$ outlines the specific requirements needed to fill those placeholders. It can be defined as:

$$d_n^{init}, \mathbb{S} = \underset{TE}{\operatorname{argmax}} \, \mathrm{P}\left(\widehat{d_n^{init}, \mathbb{S}} | \mathcal{P}_{TE}, \mathbb{H}, d_n^i\right), \tag{2}$$

where $\widehat{d_n^{init}, \mathbb{S}}$ is the potential reasoning path for $d_n^{init}, \mathbb{S}$ in the LLM. $\mathcal{P}_{TE}$ presents the prompt about the Sketch Planning Agent. $\mathbb{H} = \left\{\left(d_1^i, d_1^r\right), \left(d_2^i, d_2^r\right), ..., \left(d_m^i, d_m^r\right)\right\}$, user input d_n^i belongs to the n-turn dialogue and $n = m + 1$. $d_n^{init} = (w_1, ..., p_l, w_t)$, p_l presents the l-th placeholder, w_t presents the t-th word. $\mathbb{S} = \{s_1, s_2, ..., s_l\}$, for the l-th placeholder, it will have a corresponding retrieval request s_l in $\mathbb{S}$. For example, in the sketch response "*I recommend visiting @address*", "*@address*" is the placeholder.

3.2 Knowledge Retrieval Agent

Based on the retrieval request set $\mathbb{S}$ established by the Sketch Planning Agent, the Knowledge Retrieval Agent must extract

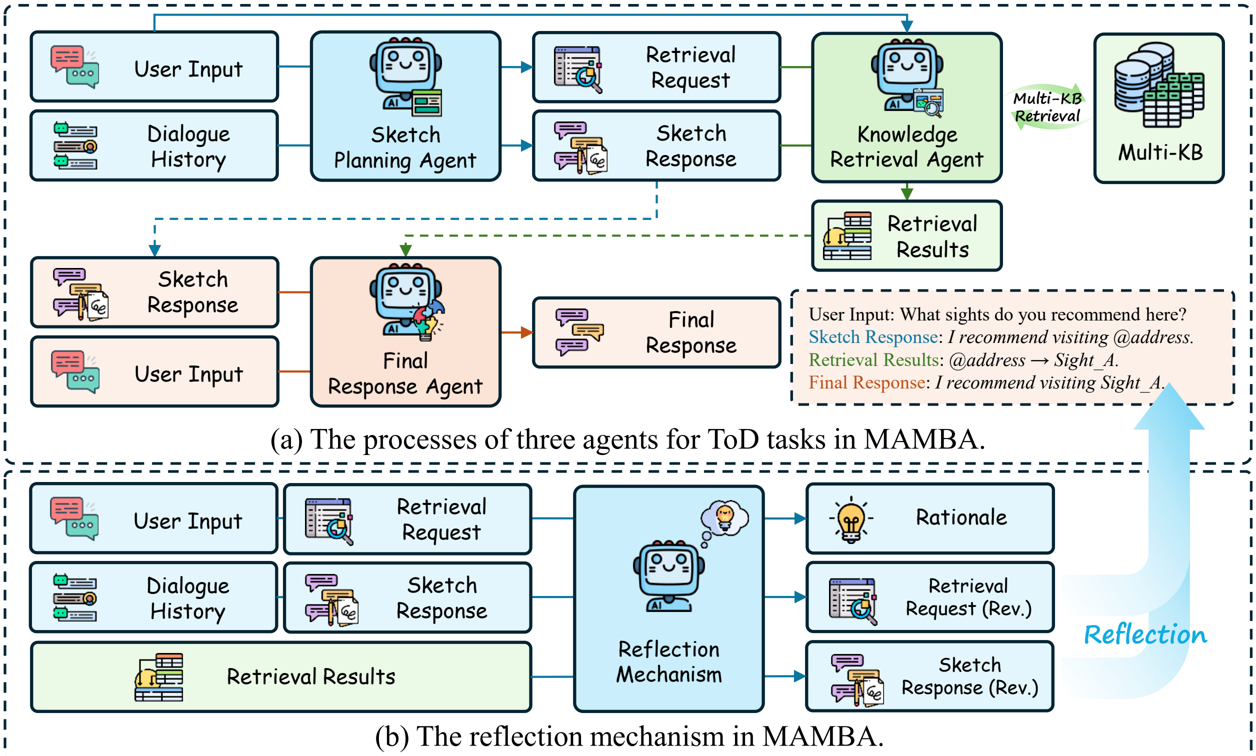


Figure 2: The main framework of MAMBA. Firstly, our framework effectively models the processes in ToD tasks through three agents (shown in Figure (a)). Subsequently, the reflection mechanism improves the performance of framework on ToD tasks (shown in Figure (b)).

retrieved results $\mathbb{R}$ from multiple knowledge bases $\mathbb{B}$. The retrieved results will correspond to the placeholder p_l in the sketch response d_n^{init}. It can be defined as:

$$\mathbb{R} = \underset{RA}{\operatorname{argmax}}\, \mathrm{P}\left(\widehat{\mathbb{R}}|\mathcal{P}_{RA}, d_n^i, d_n^{init}, \mathbb{S}, \mathbb{B}\right), \tag{3}$$

where $\widehat{\mathbb{R}}$ is the potential reasoning path for $\mathbb{R}$ in the LLM. $\mathcal{P}_{RA}$ presents the prompt about the Knowledge Retrieval Agent. For each p_l, there is a corresponding search result $r_l, r_l \in \mathbb{R}$.

3.3 Final Response Agent

After obtaining the sketch response d_n^{init} and the retrieved results $\mathbb{R}$, the Final Response Agent needs to integrate these elements to produce the final response d_n^{final} based on the user input. It can be defined as:

$$d_n^{final} = \underset{CA}{\operatorname{argmax}}\, \mathrm{P}\left(\widehat{d_n^{final}}|\mathcal{P}_{CA}, d_n^i, d_n^{init}, \mathbb{R}\right), \tag{4}$$

where $\widehat{d_n^{final}}$ is the potential reasoning path for d_n^{final} in the LLM. $\mathcal{P}_{CA}$ presents the prompt about the Final Response Agent. d_n^{final} presents the final response in MAMBA.

3.4 Reflection Mechanism

To further improve the performance of the framework, we propose a reflection mechanism. Specifically, we introduce a reflection thinking agent that re-evaluates and reflects on the results produced by the three previously mentioned agents. After obtaining d_n^{init} and $\mathbb{R}$, the reflection thinking agent will carefully evaluate these results in relation to the specific context of the dialogue $\left(d_m^i, d_m^r\right), m = n-1$ and the user input d_n^i, ultimately providing a judgment based on this assessment. If the reflection thinking agent determines that no changes are necessary, it retains the results from the initial round. However, if changes are required, it provides the revision rationale $\mathcal{R}_{rev}$, the revised sketch response $d_n^{init\prime}$ and the updated retrieval request $\mathbb{S}'$. In mathematics, if the modification is required, it can be defined as:

$$\begin{aligned}\mathcal{R}_{rev}, d_n^{init\prime}, \mathbb{S}' &= \underset{RT}{\operatorname{argmax}}\, \mathrm{P}(\widehat{\mathcal{R}_{rev}, d_n^{init\prime}, \mathbb{S}'}|\mathcal{P}_{RT}, \\ &\quad \left(d_m^i, d_m^r\right), d_n^i, d_n^{init}, \mathbb{S}, \mathbb{R}),\end{aligned} \tag{5}$$

where $\widehat{\mathcal{R}_{rev}, d_n^{init\prime}, \mathbb{S}'}$ is the potential reasoning path for $\mathcal{R}_{rev}, d_n^{init\prime}, \mathbb{S}'$ in the LLM. $\mathcal{P}_{RT}$ presents the prompt about the reflection thinking agent.

Once modifications are made, our framework will proceed with retrieval and response generation based on the revised content. It can be defined as:

$$\begin{aligned}\mathbb{R}' &= \underset{RA}{\operatorname{argmax}}\, \mathrm{P}\left(\widehat{\mathbb{R}'}|\mathcal{P}_{RA}, d_n^i, d_n^{init\prime}, \mathbb{S}', \mathbb{B}\right), \\ d_n^{final\prime} &= \underset{CA}{\operatorname{argmax}}\, \mathrm{P}\left(\widehat{d_n^{final\prime}}|\mathcal{P}_{CA}, d_n^i, d_n^{init\prime}, \mathbb{R}'\right),\end{aligned} \tag{6}$$

where $\mathbb{R}'$ presents the updated retrieval results, $d_n^{final'}$ presents the revised final response.

4 Experiments

4.1 Experimental Settings

Following previous work [Qin *et al.*, 2023a], we use the multi-KB dialogue dataset by Qin *et al.* [2023a] based on CrossWOZ [Zhu *et al.*, 2020] and RiSAWOZ [Quan *et al.*, 2020]. We use the test set for our LLM-based evaluation. We use the F1 score as the evaluation metric.

We conduct experiments on five LLMs: Llama-3.1-8B-Instruct [Grattafiori *et al.*, 2024], Gemma-3-12B-It [Team *et al.*, 2025], GPT-3.5-Turbo [Brown *et al.*, 2020], Qwen-3-8B [Yang *et al.*, 2025], and GPT-4o-mini [Hurst *et al.*, 2024]. All LLMs in are non-thinking mode. We use the official settings for all models. In Qwen series models, temperature is 0.7, and top-p is 0.8. In other models, temperature is 1.0, and top-p is 1.0.

4.2 Baselines

Following Qin *et al.* [2023a]'s work, for end-to-end ToDs, we select seven frameworks: Seq2Seq+Attn [Eric *et al.*, 2017], Mem2Seq [Madotto *et al.*, 2018], GLMP [Wu *et al.*, 2019], DDMN [Wang *et al.*, 2020], DF-Net [Qin *et al.*, 2020], Neuro-Symbolic RF [Yang *et al.*, 2022], KoK-HAN [Qin *et al.*, 2023a]. All frameworks were trained on the training set.

For LLM-based methods, we select four methods including Zero-shot Chain-of-Thought (CoT) [Kojima *et al.*, 2022], Self-Refine [Madaan *et al.*, 2023], Wrong-of-Thought (WoT) [Zhang *et al.*, 2024], S^3 Agent [Wang *et al.*, 2025c]. We reimplement these methods to adapt them for multi-KB ToD tasks. The introduction to these methods is as follows:

- Zero-shot CoT [Kojima *et al.*, 2022]: Zero-shot CoT enhances the reasoning capabilities of large language models by employing the prompt "Let's think step by step!" Given that the dataset is in Chinese, we use the same prompt for Chinese.
- Self-Refine [Madaan *et al.*, 2023]: Self-Refine first generates an initial response by the LLM, then iteratively refines the initial output through further evaluations.
- WoT [Zhang *et al.*, 2024]: WoT first generates an initial output and then conducts self-verification from three perspectives. It utilizes wrong information to guide the next output generation of LLMs.
- S^3 Agent [Wang *et al.*, 2025c]: S^3 Agent collaborates from three perspectives: superficial expression, semantic information, and sentiment expression, and combines these perspectives to obtain the final output.

4.3 Main Results

The main results are shown in Table 1. Based on these results, we observe the following findings:

(1) Existing general LLM-based methods face challenges when dealing with multi-KB ToD tasks. Essentially, all LLM baselines tend to stabilize within a limited range of performance, suggesting inherent difficulties in adapting to the complexity of multi-KB ToD tasks. Notably, their performance fails to surpass even early supervised methods such as DDMN and DF-Net. Surprisingly, the LLM baselines performed similarly on both datasets and did not attain higher F1 scores on RiSAWOZ compared to CrossWOZ, which is different from EToDs and our method. Analysis of the logs indicated that general LLM-based methods predict a greater number of slots, which negatively impacts the performance of LLMs across all datasets.

Method	CrossWOZ-F1	RiSAWOZ-F1
EToDs		
Seq2Seq+Attn [Eric *et al.*, 2017]	7.53	10.30
Mem2Seq [Madotto *et al.*, 2018]	11.33	21.83
GLMP [Wu *et al.*, 2019]	19.12	30.56
DDMN [Wang *et al.*, 2020]	29.23	29.71
DF-Net [Qin *et al.*, 2020]	29.57	34.51
Neuro-Symbolic RF [Yang *et al.*, 2022]	30.25	40.76
KoK-HAN [Qin *et al.*, 2023a]	38.07	50.87
Llama-3.1-8B-Instruct		
Zero-shot CoT [Kojima *et al.*, 2022]	20.26	18.68
Self-Refine [Madaan *et al.*, 2023]	14.45	6.88
WoT [Zhang *et al.*, 2024]	**20.64**	16.91
S^3 Agent [Wang *et al.*, 2025c]	18.35	19.01
MAMBA	10.32	**23.20**
Gemma-3-12B-It		
Zero-shot CoT [Kojima *et al.*, 2022]	21.41	22.90
Self-Refine [Madaan *et al.*, 2023]	21.77	21.76
WoT [Zhang *et al.*, 2024]	**26.76**	25.81
S^3 Agent [Wang *et al.*, 2025c]	26.16	23.06
MAMBA	18.42	**40.68**
GPT-3.5-Turbo		
Zero-shot CoT [Kojima *et al.*, 2022]	29.95	24.85
Self-Refine [Madaan *et al.*, 2023]	26.77	23.50
WoT [Zhang *et al.*, 2024]	29.04	25.33
S^3 Agent [Wang *et al.*, 2025c]	22.51	20.23
MAMBA	**30.88**	**51.75**
Qwen-3-8B		
Zero-shot CoT [Kojima *et al.*, 2022]	32.37	30.28
Self-Refine [Madaan *et al.*, 2023]	32.24	29.05
WoT [Zhang *et al.*, 2024]	30.90	28.74
S^3 Agent [Wang *et al.*, 2025c]	27.06	23.62
MAMBA	**33.34**	**47.63**
GPT-4o-mini		
Zero-shot CoT [Kojima *et al.*, 2022]	30.31	25.70
Self-Refine [Madaan *et al.*, 2023]	22.86	22.46
WoT [Zhang *et al.*, 2024]	29.03	26.43
S^3 Agent [Wang *et al.*, 2025c]	27.85	22.32
MAMBA	**40.00**	**51.27**

Table 1: Main results of EToDs and LLM-based methods in CrossWOZ and RiSAWOZ. The **bold number** denotes the best performance of method on the current LLM. The underlined number denotes the second best performance of method on the current LLM.

(2) General frameworks struggle to achieve optimal performance on multi-KB ToD tasks. Compared to zero-shot CoT, most baselines fail to surpass zero-shot CoT, even when we re-implement these frameworks specifically for multi-KB ToD tasks. This underscores that without a specialized architecture designed to model multi-KB ToD tasks, it is difficult to fully leverage the capabilities of LLMs on this task. Our framework effectively models this task and also effectively

Method	CrossWOZ-F1	RiSAWOZ-F1
Llama-3.1-8B-Instruct		
Self-Refine (initial response)	19.43	9.72
Self-Refine (refine response)	14.45 ▼	6.88 ▼
MAMBA (before reflection)	11.99	18.60
MAMBA (after reflection)	10.32 ▼	23.20 ▲
Gemma-3-12B-It		
Self-Refine (initial response)	23.44	23.19
Self-Refine (refine response)	21.77 ▼	21.76 ▼
MAMBA (before reflection)	20.28	39.30
MAMBA (after reflection)	18.42 ▼	40.68 ▲
GPT-3.5-Turbo		
Self-Refine (initial response)	30.18	25.13
Self-Refine (refine response)	26.77 ▼	23.50 ▼
MAMBA (before reflection)	25.26	48.31
MAMBA (after reflection)	30.88 ▲	51.75 ▲
Qwen-3-8B		
Self-Refine (initial response)	32.02	30.28
Self-Refine (refine response)	32.24 ▲	29.05 ▼
MAMBA (before reflection)	30.05	45.68
MAMBA (after reflection)	33.34 ▲	47.63 ▲
GPT-4o-mini		
Self-Refine (initial response)	30.73	25.99
Self-Refine (refine response)	22.86 ▼	22.46 ▼
MAMBA (before reflection)	39.26	48.74
MAMBA (after reflection)	40.00 ▲	51.27 ▲

Table 2: The performance of Self-Refine and MAMBA. The downward arrow ▼ indicates that refinement does not improve performance in Self-Refine/MAMBA. An upward arrow ▲ indicates that refinement can further improve performance of Self-Refine/MAMBA.

Dimensions	Description
Fluency	Are the responses grammatically correct and clearly articulated? Are there any awkward or stiff phrasings?
Coherence	Is the logic consistent across multiple turns? Does the assistant correctly understand and maintain the context?
Naturalness	Do the responses sound like natural human conversation? Is the tone friendly and appropriate?
Criteria	**Description**
0.0	Completely unintelligible / Severely disjointed
1.0	Mostly disjointed
2.0	Significant issues but basically understandable
3.0	Generally fluent with minor issues
4.0	Fluent and natural with occasional slight flaws
5.0	Perfectly fluent and natural

Table 3: Details of evaluation dimensions and criteria in the LLM evaluation for final response.

Model	CrossWOZ	RiSAWOZ
Llama-3.1-8B-Instruct	2.00	2.55
Gemma-3-12B-It	3.60	3.94
GPT-3.5-Turbo	3.92	4.39
Qwen-3-8B	3.50	3.88
GPT-4o-mini	3.78	3.99

Table 4: The average score for each dialogue after the LLM evaluation in CrossWOZ and RiSAWOZ.

mitigates the problem of slot redundancy.

(3) Our framework achieves superior performance across diverse backbones. Experimental results show that our framework achieves the best performance on most models, demonstrating its effectiveness in multi-KB ToD tasks. More importantly, our method outperforms the current state-of-the-art framework KoK-HAN without training. Even with a smaller LLM like Qwen-3-8B, our method significantly outperforms LLM-based methods and achieves competitive performance compared to EToDs.

5 Analysis

To better understand the details of our framework in multi-KB ToD tasks, we conduct further analysis as follows.

5.1 The reflection mechanism in MAMBA can effectively enhance the performance of LLMs in multi-KB ToD tasks

To validate the efficacy of our reflection mechanism, we conduct a comparative analysis of model performance across varying backbones before and after the reflection stage, as detailed in Table 2.

As observed, the integration of the reflection mechanism in MAMBA yields consistent performance gains across the majority of models, substantiating the effectiveness of the reflection mechanism in MAMBA for multi-KB ToD tasks. In contrast, one of the baseline methods, self-refine, has difficulty enhancing performance. In most models, the performance after refinement is even worse than the performance of the initial response in Self-Refine. This disparity underscores the critical importance of precise task modeling; our framework provides the necessary structural grounding for the dialogue task, thereby enabling the system to fully leverage the potential of self-iteration, a capability that generic frameworks fail to unlock.

5.2 MAMBA ensures accuracy while generating fluent and natural responses

To assess the linguistic quality of MAMBA, we leverage GPT-4o as an automated evaluator to score the responses. Following the methodology in Li *et al.* [2026], we employ a 0-5 scoring scale with 0.5 increments, assessing three dimensions: **Fluency**, **Coherence**, and **Naturalness**. The details of evaluation dimensions and criteria are delineated in Table 3.

The final results are presented in Table 4. As observed, the vast majority of the generated responses fall within the high-quality spectrum, demonstrating the effectiveness of our framework in generating satisfactory responses. This ensures that while retrieval and slot filling are executed correctly in MAMBA, the responses provided by the Final Response Agent to the user are also fluent.

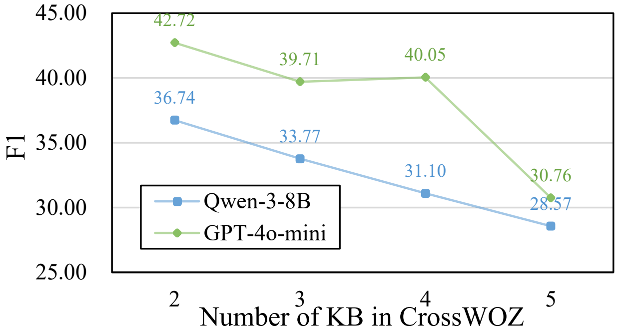
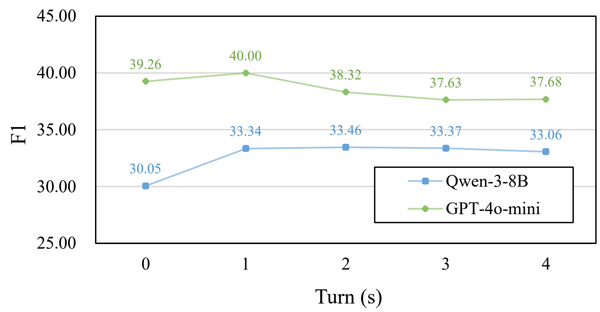
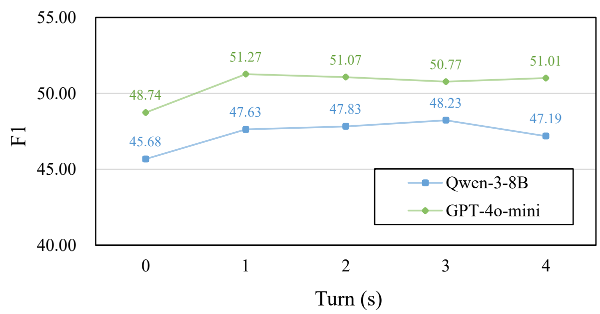


Figure 3: The performance of MAMBA in CrossWOZ with different numbers of knowledge bases.



(a) The performance of multi-turn reflection in CrossWOZ.



(b) The performance of multi-turn reflection in RiSAWOZ.

Figure 4: The performance of MAMBA across different reflection rounds.

5.3 The difficulty of ToD increases with the increasing number of knowledge bases

To investigate the correlation between task complexity and the number of knowledge bases, we analyze the performance of Qwen-3-8B and GPT-4o-mini on dialogues with varying quantities of KB in the CrossWOZ. The results are presented in Figure 3.

As illustrated, the performance of MAMBA exhibits a discernible downward trend as the number of KBs increases. This decline is likely due to the increased complexity of information retrieval and analysis associated with a higher number of KBs, aligning with the findings of previous work [Qin *et al.*, 2023a]. These findings highlight the inherent challenge of scalability in multi-KB scenarios, underscoring a critical direction for future research to enhance model robustness in multi-KB.

Method	CrossWOZ	RiSAWOZ
Ground Truth	18.13	11.25
Qwen-3-8B		
Zero-shot CoT [Kojima *et al.*, 2022]	118.41	290.06
Self-Refine [Madaan *et al.*, 2023]	118.98	304.83
WoT [Zhang *et al.*, 2024]	130.03	300.28
S^3 Agent [Wang *et al.*, 2025c]	150.24	388.87
MAMBA	17.41	15.88
GPT-4o-mini		
Zero-shot CoT [Kojima *et al.*, 2022]	117.10	319.98
Self-Refine [Madaan *et al.*, 2023]	188.59	383.17
WoT [Zhang *et al.*, 2024]	95.96	313.48
S^3 Agent [Wang *et al.*, 2025c]	144.82	302.11
MAMBA	16.20	12.19

Table 5: The statics of slots in the response for baselines and our method. The results represent the average number for each dialogue.

5.4 Excessive reflection depth can actually have a negative impact on performance

We further investigate the relationship between reflection depth and model efficacy within MAMBA by evaluating Qwen-3-8B and GPT-4o-mini across multiple reflection iterations. The final results are shown in Figure 4.

We observe a distinct phenomenon of performance saturation: the models attain optimal results after the initial reflection iteration. Subsequent rounds yield negligible marginal gains, with performance metrics remaining stable. This stability implies that for multi-KB ToD tasks, a single reflection round effectively exhausts the ability of the model to self-correct; remaining inaccuracies likely stem from the intrinsic capability limitations of large language models rather than insufficient self-correction depth.

5.5 MAMBA can prevent the decline in LLM when following instructions in long-text

To further investigate the precision of our method in slot prediction, we conduct a quantitative analysis of the generated slot counts. The comparative results are presented in Table 5.

As evidenced by the data, baseline methods exhibit a marked tendency towards slot over-prediction. This redundancy leads to the retrieval of extraneous information unrelated to the user's specific query. For instance, in response to the user query "What sights do you recommend here?", the baseline not only outputs "sight" but also related slots. We suspect this occurs because baseline methods have difficulty following prompt instructions and generating accurate short answers when dealing with long-text [Zhang *et al.*, 2025]. In contrast, our method accurately captures user intent and provides precise responses. The final number of predicted slots closely matches the ground truth, demonstrating the effectiveness of our three-stage framework for modeling ToD tasks.

5.6 Case study

To better understand the differences between our method and baselines, we conduct a case study analysis, with the results

Case 1	
Dialogue Context	Turn 1: User asks for train from Shanghai to Nanjing → System recommends D3136 Turn 2: User asks for first-class ticket price → System replies 178 yuan
KB (excerpt)	D3136: { duration: "2h42min", departure: 17:46, arrival: 20:28 }
User Input	How long does this train take?
Method	**Output**
MAMBA (before reflection)	D3136 takes about 1 hour 45 minutes.
MAMBA (after reflection)	D3136 takes about 2 hours 42 minutes.
Case 2	
Dialogue Context	Turn 4: User requests hotel recommendation → System recommends "Tongli Moonrise Hotel" Turn 5: User asks for hotel address → System replies with the hotel's address
KB (excerpt)	Tongli Moonrise Hotel: { phone: "0512-63052900", address: "No.53 Dongxi Street..." } Suzhou Museum: { phone: "0512-67575666" }
User Input	Please also look up the phone number.
Method	**Output**
Zero-shot CoT	Suzhou Museum's phone number is 0512-67575666.
MAMBA (after reflection)	The phone number for Tongli Moonrise Hotel is 0512-63052900.

Table 6: Case study of MAMBA in before/after reflection (case 1) and comparison of baseline and our method (case 2). Our method reduces hallucination in LLMs through reflection and accurately identifies user needs in multi-turn dialogue.

presented in Table 6.

In the first case, when the user asks about the train travel duration, the before reflection stage hallucinates an incorrect value "1 hour 45 minutes" that does not exist in the knowledge base. After applying our reflection mechanism, the system correctly extracts the accurate duration "2 hours 42 minutes" from the KB. In the second case, the zero-shot CoT baseline produces a different type of hallucination: when the user asks for the hotel's phone number, zero-shot CoT incorrectly returns the museum's phone number due to entity confusion across dialogue turns. In contrast, our method correctly tracks the conversation context and extracts the accurate phone number from the corresponding KB entry.

6 Related Work

Task-oriented dialogue in multiple knowledge bases requires models to accurately understand user intent and retrieve relevant content to generate responses. Early works on this task focus on developing specific frameworks. Zeng *et al.* [2022] introduce HM2Seq, it uses hierarchical memory to represent knowledge base and improving the reasoning ability of ToDs while reducing the need for manual annotations. Yang *et al.* [2022] propose a two-stage method that selects effective hypotheses to enhance dialogue performance. Qin *et al.* [2023a] introduce KoK-HAN, which improves the capabilities of system across multiple knowledge bases by modeling information from different sources. Wan *et al.* [2023] decouple retrieval and generation, enhancing performance in knowledge retrieval through precise entity retrieval and attribute filtering. As researchers increasingly focus on large language models in task-oriented dialogues [Qin *et al.*, 2023b], Chung *et al.* [2023] integrate LLMs into this field, achieving impressive results on subtasks in ToDs. Algherairy *et al.* [2025] evaluate the ability of large language models in simulating user interactions for ToDs, and find that LLMs have enormous potential in this task. Dong *et al.* [2025] effectively model task-oriented dialogue using an adaptive exploratory retrieval mechanism and a strategy planner, which improved the system's proactive engagement in conversations. Wang *et al.* [2025a] propose MAC-CIToD, employing multi-agent modeling to identify consistency in task-oriented dialogues.

Compared to previous work, we first introduce MAMBA, a multi-agent framework for modeling ToD tasks. By emulating the intrinsic workflows of ToDs through specialized agents, our framework effectively augments the reasoning capabilities of LLMs. Additionally, we incorporate a reflection mechanism that iteratively refines intermediate outputs, thereby enhancing the overall performance of the framework.

7 Conclusion

In this paper, to address the critical bottleneck of data dependency in end-to-end task-oriented dialogue (ToD) systems across multiple knowledge bases, we propose MAMBA, a novel framework that orchestrates multi-agent collaboration by modeling ToD tasks to accurately understand user intent and retrieve multi-KB, and returns the response. Crucially, we introduce a reflection mechanism into MAMBA, which further enhances the performance of large language models on ToD tasks through iterative self-feedback from large language models. Extensive experiments on CrossWOZ and RiSAWOZ demonstrate the effectiveness of our framework, which notably surpasses the state-of-the-art supervised baseline, KoK-HAN, in a zero-shot setting. We hope that our research will provide valuable insights for the future development of training-independent task-oriented dialogue systems.

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